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Pose coverage and held-out image validation for multi-camera robot calibration

Author

Xiongyule Zhou

Affiliation

School of Mechanical, Electronic and Control Engineering, Beijing Jiaotong University, Beijing 100044, China

Corresponding author

Xiongyule Zhou
Email: jensenzhou47@gmail.com
ORCID: 0009-0005-1179-9879
No. 3 Shangyuancun, Haidian District, Beijing 100044, China

Pose coverage and held-out image validation for multi-camera robot calibration

Purpose To quantify how pose selection, solver choice and pixel refinement affect held-out consistency in robot-mounted camera calibration.

Design/methodology/approach Experiments used head, body and wrist cameras, a planar tag target, 990 training and 288 validation poses, and 6,383 images across nine sessions. Four policies were compared at equal pose budgets. A matched ten-pose experiment evaluated five solvers and pixel refinement in 2,808 pipeline attempts.

Findings At ten poses, farthest-point/random median ratios under Park fitting were 0.854 for translation closure and 0.912 for image error. With identical subsets and ten random draws, the image ratio changed from 0.911 to 0.993 after pixel refinement. Farthest selection still outperformed concentrated selection in 29/36 refined comparisons. All 36 full-set pixel fits converged with positive depth and achieved a validation root-mean-square (RMS) error of 0.247–0.616 pixels. Solver failures depended on motion geometry and selection policy.

Research limitations/implications Validation poses were local and accessible to the calibration workflow. No external metrology or task-accuracy test was available.

Originality/value Matched pose budgets and explicit validity checks reveal that a geometric selection advantage can weaken after changing the fitting objective, supporting conditional calibration assessment.

Keywords Hand-eye calibration; Multi-camera robots; Pose coverage; Held-out validation; Reprojection error; Reproducibility

Article classification Research paper

1 Introduction

Robot-mounted cameras require a stable geometric relationship to their supporting links. On a platform with head, body and wrist cameras, the available calibration motions differ substantially between mounting locations. A small image residual alone does not show whether the selected robot poses sufficiently constrain the camera-to-link transform or whether the estimate predicts observations at omitted poses.

Industrial calibration is ultimately judged against manufacturing or manipulation requirements. Bai *et al.* (2020) combined calibration and detection in mobile robotic manufacturing, while Wang *et al.* (2025) evaluated robot self-calibration using a line-structured light sensor. Before relating a robot-camera estimate to such application requirements, an experimental comparison should distinguish pose selection, solver behaviour and the objective used for refinement. These choices can change an internal validation residual without establishing a corresponding change in physical positioning accuracy.

We designed and conducted multi-camera robot calibration experiments using a planar fiducial target and separate training and validation pose groups. The experiments cover four camera roles, nine complete sessions and eight acquisition groups. We compare four pose-selection policies at equal training-pose budgets, five hand-eye solvers, and training-only pose or pixel refinement. All policies use the same measured candidate poses and validation observations; selecting a subset does not create an additional robot trial.

The study asks when geometric coverage improves held-pose and held-image consistency. It tests whether that improvement survives changes in solver or fitting objective and defines the acquisition conditions supporting the measured performance. Its contribution is a matched experimental evaluation: 6,383 accepted images, 14,832 primary subset fits, and a 2,808-attempt ten-pose comparison that separates solver failures from finite but physically invalid projections. The result is a conditional assessment of calibration practice rather than a new solver or a claim of absolute positioning accuracy.

2 Related work

Classical hand-eye calibration estimates a rigid relationship from paired robot and camera motions. Tsai and Lenz (1989) separated rotation and translation estimation, while Park and Martin (1994) used the Euclidean group. Horaud and Dornaika (1995) examined alternative formulations, and Daniilidis (1999) coupled the rigid-motion components through dual quaternions. Andreff *et al.* (1999) developed a linear formulation for hand-eye calibration. Enebuse *et al.* (2022) demonstrated the influence of noise and motion range on estimation accuracy. These studies motivate reporting motion geometry and implementation-specific failures alongside residuals.

Pose design can improve the information supplied to calibration. Wu *et al.* (2015) studied measurement configurations for robot geometric calibration. Yang *et al.* (2023) selected eye-in-hand views using predicted uncertainty reduction, and Li *et al.* (2026) investigated reinforcement-learning-based next-best-pose calibration. Our geometric policies select from a common measured candidate set without predicting image noise or generating new robot motions. They provide controlled budget comparisons, rather than reproductions of those active methods.

The fitting objective also affects interpretation. Tabb and Ahmad Yousef (2017) compared iterative robot-world/hand-eye formulations using multiple metrics. Vieira *et al.* (2026) jointly addressed extrinsic and kinematic calibration with multiple patterns. We keep kinematics and camera intrinsics fixed and compare pose closure with image reprojection. This isolates the effect of the fitted extrinsic model while leaving systematic kinematic, intrinsic and target errors within both endpoints.

Multi-camera platforms additionally raise the question of inter-camera registration. Allegro *et al.* (2024) used shared target-to-end-effector and camera-to-camera constraints in industrial workcells. Our four camera roles were measured in separate sequences on different support links. The experiments therefore evaluate camera-to-support calibrations within a multi-camera robot; they do not estimate or validate a synchronized camera network.

3 Experimental apparatus and methods

The apparatus, acquisition groups and evaluation units are specified below. The validation pose identifiers were separate during collection, but their results were accessible to the calibration workflow. The study was not preregistered and does not treat these observations as a previously untouched blind test. Primary analysis settings were fixed before evaluating the six additional acquisition groups; the matched solver/objective comparison is a secondary analysis.

3.1 Robot, cameras and calibration target

The robot platform family has two seven-joint arms, a four-joint trunk and a two-joint head. Four color-camera roles were calibrated: head, body, left wrist and right wrist. Their support frames are Head2, cam_install_body, base_L_wrist and base_R_wrist, respectively. A camera-to-support transform was fitted separately for each role and session. Figure 1 defines the apparatus relationships and measurement flow; Plate 1 shows actual wrist-camera target views.

Images have a resolution of 640 × 480 pixels. Each sequence uses its camera matrix K and five-coefficient plumb-bob distortion vector D, with no additional image undistortion. Across the main sessions, horizontal focal lengths are 602.1–609.9 pixels for the head, 387.2–389.2 pixels for the body and 390.6–395.1 pixels for the wrists. Camera-to-target-origin distances estimated from image poses span 0.988–1.154 m, 0.632–0.929 m and 0.741–0.887 m, respectively. These distances describe the modelled viewing geometry, not an independent range measurement.

The planar target has seven columns and five rows of tags, a nominal tag side of 35 mm and a spacing ratio of 0.3. The resulting gap is 10.5 mm, pitch 45.5 mm and patterned extent 308 × 217 mm excluding margins. Tags use the legacy tag36h11 payload with two black border cells; identifiers 0–34 follow row-major order. Target scale and flatness were not independently measured. Stating these quantities is essential because target configuration determines both metric pose scale and corner correspondence (ETH Zurich ASL, n.d.).

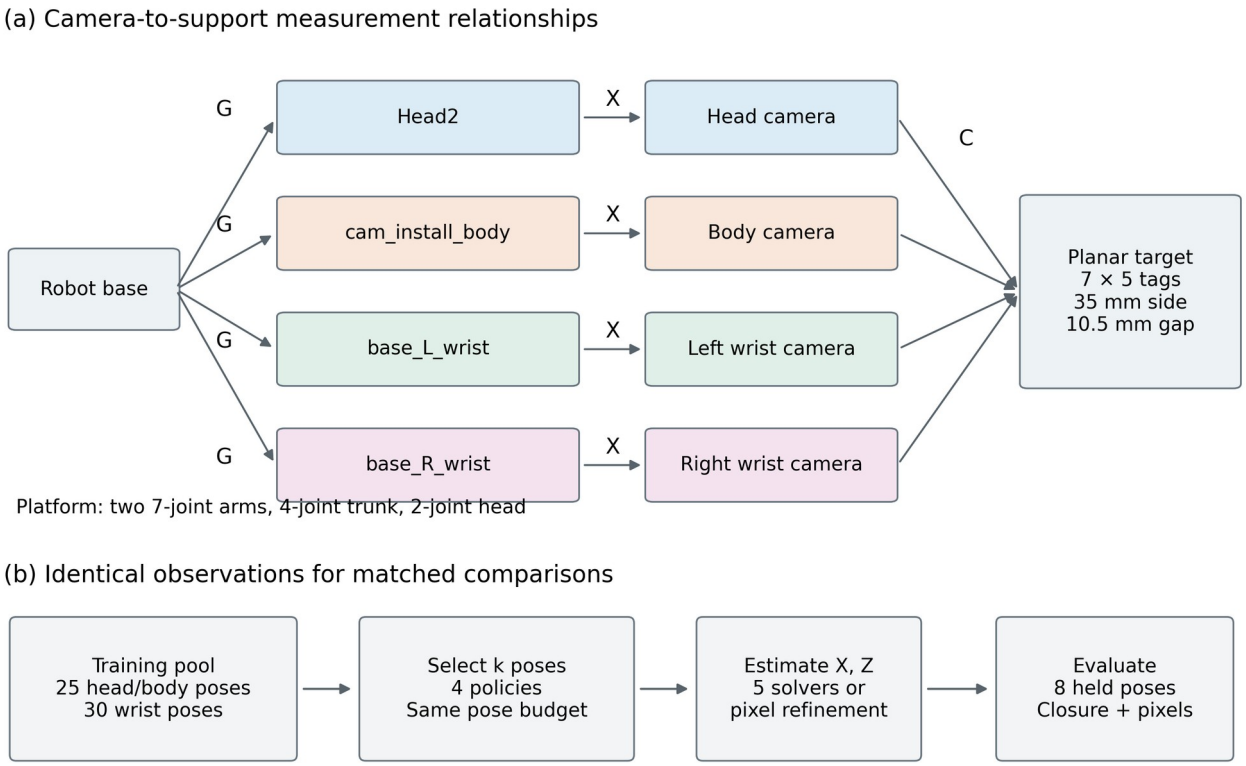


Figure 1. Experimental apparatus relationships and evaluation flow. Each camera is related to its own support link, with the planar target fixed within a role sequence. The diagram is a transform/measurement schematic, not a dimensional robot drawing. The same training pool and eight validation poses are used for each matched pipeline. Source: author's experimental configuration.

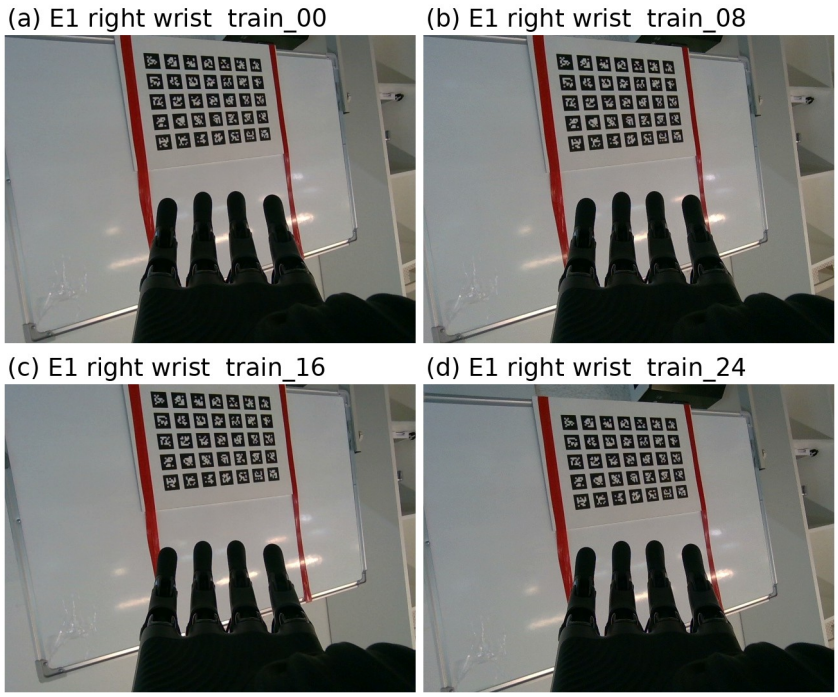


Plate 1. Four original wrist-camera target views from fixed training-pose identifiers. Each 640 × 480 image is retained at its original pixel dimensions within a labelled 300 dpi composite; the panel does not imply synchronized acquisition. Source: author's experimental images.

3.2 Pose arrangement, acquisition and validation split

We collected head, body and both wrist sequences in each of nine complete sessions during September–October 2026 (Table I). Acquisition groups A–H identify the platform/session groupings; A contributes two dates. Seven groups retain a robot identifier in their session metadata. Group D lacks that identifier and is included with an explicit exclusion sensitivity. Consequently, eight groups should not be interpreted as eight independently verified hardware replicates.

Each head or body sequence contains 25 training poses, each wrist sequence 30, and every sequence eight validation poses. The target provides a fixed reference within a role sequence; its base-frame pose is estimated separately for each fit. The acquisition used nominally five frames per pose. Of 6,390 candidate frames at 1,278 pose groups, 6,383 were accepted: 4,944 training and 1,439 validation images. There are 990 training and 288 validation pose groups. A total of 1,273 pose groups have five accepted frames, three have four and two have three. The seven rejected candidates are retained in the inclusion record. This quality-screened sample does not estimate detection reliability over a continuous recording.

The measured pose arrangements differ by role (Figure 2). Maximum pairwise support-link rotation spans are 22.34–27.28° for the head, 13.41–13.42° for the body, 5.42–5.88° for the left wrist and 5.35–5.67° for the right wrist. Corresponding maximum pairwise support translations are 8.86 mm, 187.35–188.68 mm, 19.65–22.27 mm and 17.21–18.73 mm. The rotations and translations describe achieved poses; the study does not assume a uniform angular grid or a factorial manipulation of motion axes.

Training and validation are separated by pose identifier, so repeated images from one pose do not enter both sets. Validation remains local: nearest-training separations are 0.23–3.54° and 1.11–13.66 mm. Additional analyses use 23 incomplete four-role or additional single-role sequences (3,811 images), and five further head sequences (825 images). These groups are reported separately and do not enlarge the main matched-session denominator.

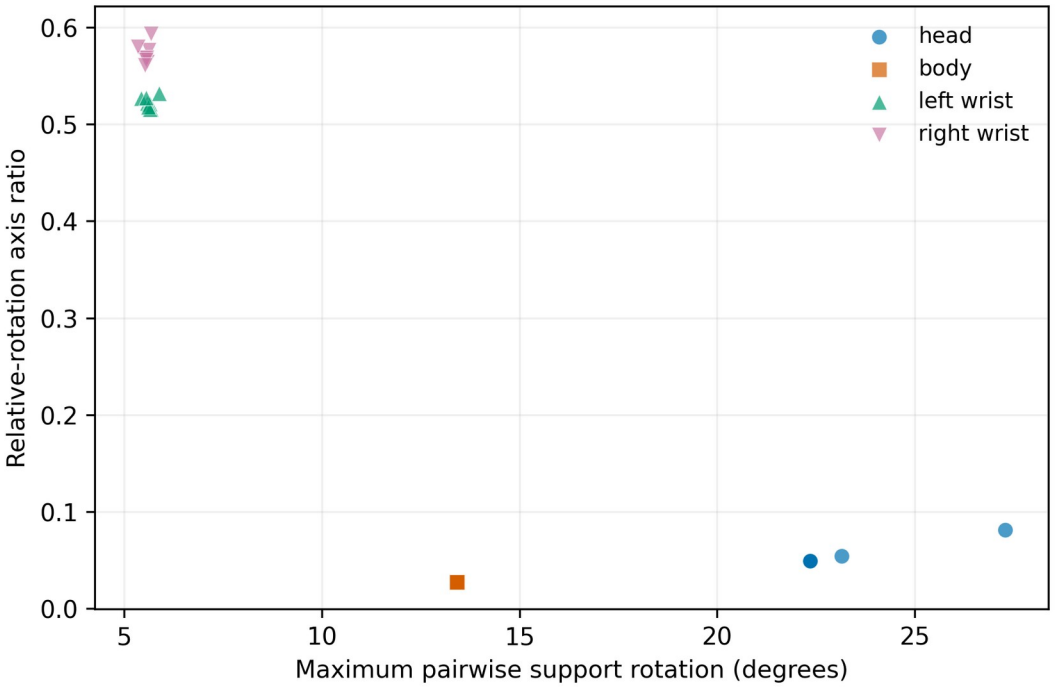


Figure 2. Achieved training-pose geometry in the 36 main camera-session panels. The relative-rotation axis ratio is the smallest-to-largest singular-value ratio of rotation vectors. Overlapping points indicate related motion patterns; the plot does not separate role, mechanism and trajectory effects. Source: author’s measurements and calculations.

Table I. Acquisition sessions and median held-pose/image outcomes across the four camera roles. Every session has 110 training and 32 validation poses; medians do not measure whole-robot accuracy. Source: author's measurements and calculations.

Session	Date 2026	Images train/held	Park closure mm	Park image px	Pixel fit image px
A1	10 Sep	550/159	0.730	0.420	0.333
A2	20 Sep	549/160	0.526	0.419	0.375
B1	16 Oct	550/160	0.808	0.496	0.356
C1	10 Sep	550/160	0.658	0.422	0.314
D1	1 Sep	550/160	0.462	0.401	0.338
E1	8 Sep	550/160	0.755	0.491	0.349
F1	21 Sep	545/160	0.600	0.430	0.333
G1	21 Sep	550/160	0.642	0.410	0.343
H1	17 Oct	550/160	0.639	0.483	0.364

3.3 Image measurements and transform model

Corner extraction uses scale factors 1, 1.5, 2, 2.5 and 4, a maximum Hamming distance of two, a minimum tag area of 70 pixels squared and edge-based corner refinement. Iterative OpenCV perspective-n-point (PnP) estimation uses the fixed camera matrix, distortion coefficients and stated target coordinates. The accepted main images yield 22–35 tags each. The same detector and corner convention are used for every pipeline. No image is excluded because its extracted tag count differs between processing outputs.

Robot support transforms are paired to camera-header timestamps using the nearest transform sample within 200 ms. This association threshold is not a bound on exposure synchronization error. Transforms map child coordinates into parent coordinates. G_i maps the camera-support link into the robot base, and C_i maps the target into the camera. X maps the camera into its support link, and Z maps the target into the base. With the target fixed within a sequence,

$$G_i X C_i = Z. \qquad (1)$$

Translations are in metres. Within each training or validation pose, translations are averaged arithmetically and rotations by the chordal mean, separately for G and C. Poses receive equal weight regardless of their accepted frame count. Eliminating Z gives $A_{ij} = G_j^{-1} G_i$ and $B_{ij} = C_j C_i^{-1}$, satisfying $A_{ij}X = XB_{ij}$. The primary estimator is OpenCV 4.11.0 Park; Z is then the mean of training $G_i X C_i$ only. Validation observations never update X or Z.

3.4 Equal-budget pose-selection experiments

At budgets of six, ten, 15 and 20 training poses, all policies draw from the same 25- or 30-pose candidate set. Farthest-point selection starts at the pose farthest from the training medoid and repeatedly maximizes distance to the nearest selected pose. Concentrated selection takes the k poses nearest the medoid. Chronological selection uses the first k acquired poses. Random selection draws k poses without replacement, with 100 draws per panel and budget. Selected poses are fitted in acquisition order; ties are resolved in that order.

$$d(i,j) = \sqrt{\left[\left(\theta_{ij} / \ 10^\circ\right)^2 + \left(\lVert t_i - t_j \rVert \ / \ 0.1 \ m\right)^2\right]}. \qquad (2)$$

Equation (2) uses training robot geometry only. Its angular and translational scales are fixed analysis choices; they are not tuned against validation errors. Maximum rotation span, translation span and the smallest-to-largest singular-value ratio of relative rotation vectors describe the achieved excitation. The axis ratio is not a complete identifiability score.

There are 36 × 4 × 103 = 14,832 primary Park subset attempts. Random membership is fixed and saved; additional-group random streams use seed 20261019 plus a deterministic panel-identifier offset. Policies are compared within panels using farthest-point error divided by the median error of 100 random draws. Ratios

below one favour farthest-point selection. The 12 panels from A/B and 24 from C–H are also reported separately.

3.5 Solvers, refinement and matched ten-pose comparison

Full candidate sets are fitted with the Tsai, Park, Horaud, Andreff and Daniilidis options of OpenCV 4.11.0 (OpenCV, 2025a). Nonfinite transforms, invalid rotations and the Tsai identity/zero sentinel are counted as numerical failures. A pose-closure refinement jointly adjusts the 12 rigid-transform parameters of X and Z using a soft-L1 loss. It uses scales of 0.01 m and one degree, at most 200 evaluations and tolerances of 10^{-10}.

The pixel refinement also jointly adjusts X and Z, initialized by Park, but minimizes training corner reprojection error. It uses fixed intrinsics, a linear least-squares loss, equal pose weights and Jacobian-based parameter scaling. SciPy least_squares uses at most 120 evaluations and ftol, xtol and gtol of 10^{-9} (Virtanen *et al.*, 2020). No validation observations select optimizer settings or update the model. Pose closure and pixel fitting are alternative objectives, not independent reference measurements.

To compare solver and objective effects under exactly matched pose budgets, a secondary experiment uses the ten-pose farthest, concentrated and chronological subsets, and the first ten saved random draws. These 13 subsets per panel give 468 unique measured subsets. Each is fitted with five solvers and Park-initialized pixel refinement, giving 2,808 pipeline attempts. No subset is selected by its error. All six pipelines receive the same pose identifiers, camera model, corners and eight validation groups.

The matched comparison distinguishes numerical finiteness, optimizer convergence and positive target-corner depth in both training and validation images. A valid result requires all three. Nonlinear refinement depends on Park initialization; an unavailable initializer is retained as a pipeline failure. Valid-only medians always accompany their denominators, because methods can succeed on different subsets. The farthest/random objective comparison uses the same ten random draws under each objective and therefore differs from the primary 100-draw baseline.

3.6 Evaluation, dependence and sensitivity

For held pose v, $Q_v = G_v X C_v$ is the implied target-to-base transform. Translation and rotation closure relative to the training-only Z are

$$e_t(v) = 1000 \, \| t(Q_v) - t(Z) \|; e_r(v) = angle[\, R(Z)^T R(Q_v) \,]. \qquad (3)$$

Root-mean-square (RMS) error over the eight poses is reported in millimetres and degrees. For held-image error, target corners are projected through $(G_i X)^{-1} Z$ using each frame's robot transform and fixed camera model. Squared two-dimensional Euclidean corner errors are averaged within frames, then within poses, then across the eight poses, before taking the square root. Thus each pose has equal weight. The projection implementation agrees with OpenCV projectPoints to less than 10^{-8} pixels.

Conditional 95% percentile intervals use 5,000 resamples of the eight validation poses. They describe variation within that empirical pose distribution, excluding systematic errors, training uncertainty and between-robot variability. No population-level significance tests are applied to dependent panels or random draws. Acquisition-group summaries first geometrically average ratios within a group and then give equal weight to groups, so the two dates from A do not double its weight.

Sensitivity analyses omit each training pose in turn, vary five geometric scale combinations, exclude group D, and evaluate the separate additional-role sequences. Transfer tests fix a source pixel-refined X and fit only destination Z using destination training images. Eight cross-date comparisons cover A1/A2; 52 directed same-day comparisons require matching camera, target and robot-description parameters. Shared captures make these directed pairs dependent. Configuration matching does not establish unchanged physical mounting.

4 Results

4.1 Full-set calibration and attained image consistency

All 36 full-set Park estimates are finite. Held-pose translation RMS is 0.256–5.072 mm and rotation RMS 0.043–0.402°. Full-set held-image RMS is 0.278–2.732 pixels. Training-only pose-closure refinement yields 0.202–2.507 mm translation RMS; pixel refinement yields 0.247–0.616 pixels, with convergence and positive target depth in all 36 panels (Figure 5a). Table I gives session-level medians, with individual role results in the supplement.

The full-set comparison produces 27/36 numerical failures for Tsai and 9/36 for Andreff (Table II). Successful-fit medians consequently condition on different panels. OpenCV 4.11.0 excludes Tsai motion pairs when either scaled minimal-quaternion norm lies outside 0.3–1.7; fewer than two retained pairs produces no estimate (OpenCV, 2025b). The lower cutoff corresponds to approximately 17.3° rotation. Body and wrist rotation spans are below this implementation threshold. These outcomes establish a limitation of the tested implementation under the measured motions, not universal failure of the underlying method.

Table II. Solver outputs on full candidate sets and identical ten-pose subsets. Full-set medians condition on finite outputs; ten-pose validity additionally requires positive training/validation depth. Different successful subsets preclude a universal accuracy ranking. Source: author’s calculations.

Solver	Full finite	Full median mm	Full median px	Ten-pose finite	Ten-pose valid
Tsai	9/36	5.050	2.555	96/468	92/468
Park	36/36	0.647	0.443	460/468	459/468
Horaud	36/36	0.647	0.443	468/468	459/468
Andreff	27/36	1.474	0.624	351/468	344/468
Daniilidis	36/36	0.632	0.436	465/468	462/468

4.2 Pose selection at a fixed budget

At ten poses, farthest-point selection gives lower translation closure than the 100-draw random median in 9/12 A/B panels and 17/24 C–H panels (Figure 3; Table III). Median ratios are 0.833 and 0.879, respectively; C–H panel ratios range from 0.475 to 1.779. For held-image error, 17/24 C–H panels favour farthest-point selection, with a median ratio of 0.879. Across all 36 panels, median closure and image ratios are 0.854 and 0.912. These are within-panel comparisons; they do not imply that every camera benefits.

The comparison varies with budget (Figure 4). Errors do not decrease monotonically in every panel, and a subset can outperform the full candidate set. Across the four budgets, 49/14,832 primary subset attempts are numerically unsuccessful. One additional finite ten-pose chronological Park estimate predicts negative target depth and is separately flagged. Numerical failure rates therefore understate the number of unusable projections.

The 23 additional-role sequences give 20/23 favourable closure comparisons and 19/23 favourable image comparisons at ten poses; both median ratios are 0.885. Excluding group D from the main sample gives 23/32 favourable closure and 22/32 favourable image comparisons, with median ratios of 0.854 and 0.912. These checks preserve the direction of the aggregate Park comparison without providing independent hardware replication.

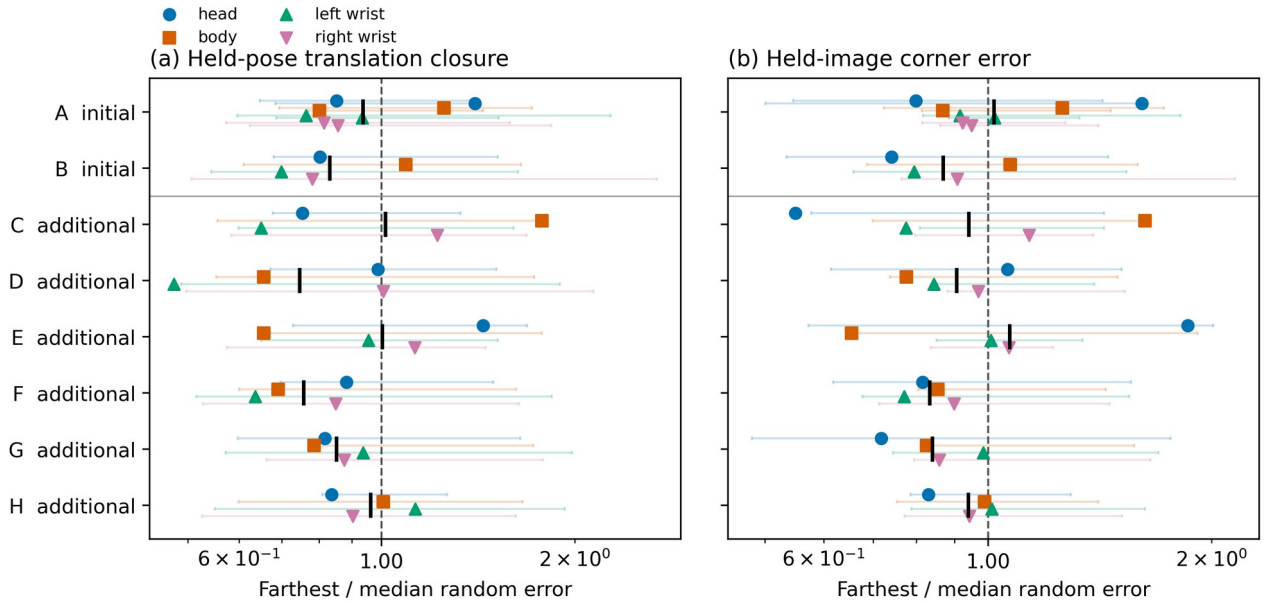


Figure 3. Ten-pose farthest-point error relative to the median of 100 random subsets. Coloured points identify camera roles and black ticks give acquisition-group geometric means. Faint horizontal bars show random-error 2.5th–97.5th percentiles normalized by their median, not confidence intervals for a policy effect. A contributes two dates. Source: author's measurements and calculations.

Table III. Ten-pose farthest/random ratios using 100 random subsets per panel. Group-balanced ratios geometrically average within acquisition group before averaging groups equally. Ratios below one favour farthest selection. Source: author's calculations.

Group	Closure < 1	Median ratio	Image < 1	Median ratio	Group ratios closure / image
A/B	9/12	0.833	8/12	0.920	0.882 / 0.941
C–H	17/24	0.879	17/24	0.879	0.883 / 0.919
All main	26/36	0.854	25/36	0.912	0.883 / 0.924
Additional roles	20/23	0.885	19/23	0.885	0.892 / 0.915
Exclude D	23/32	0.854	22/32	0.912	0.904 / 0.927

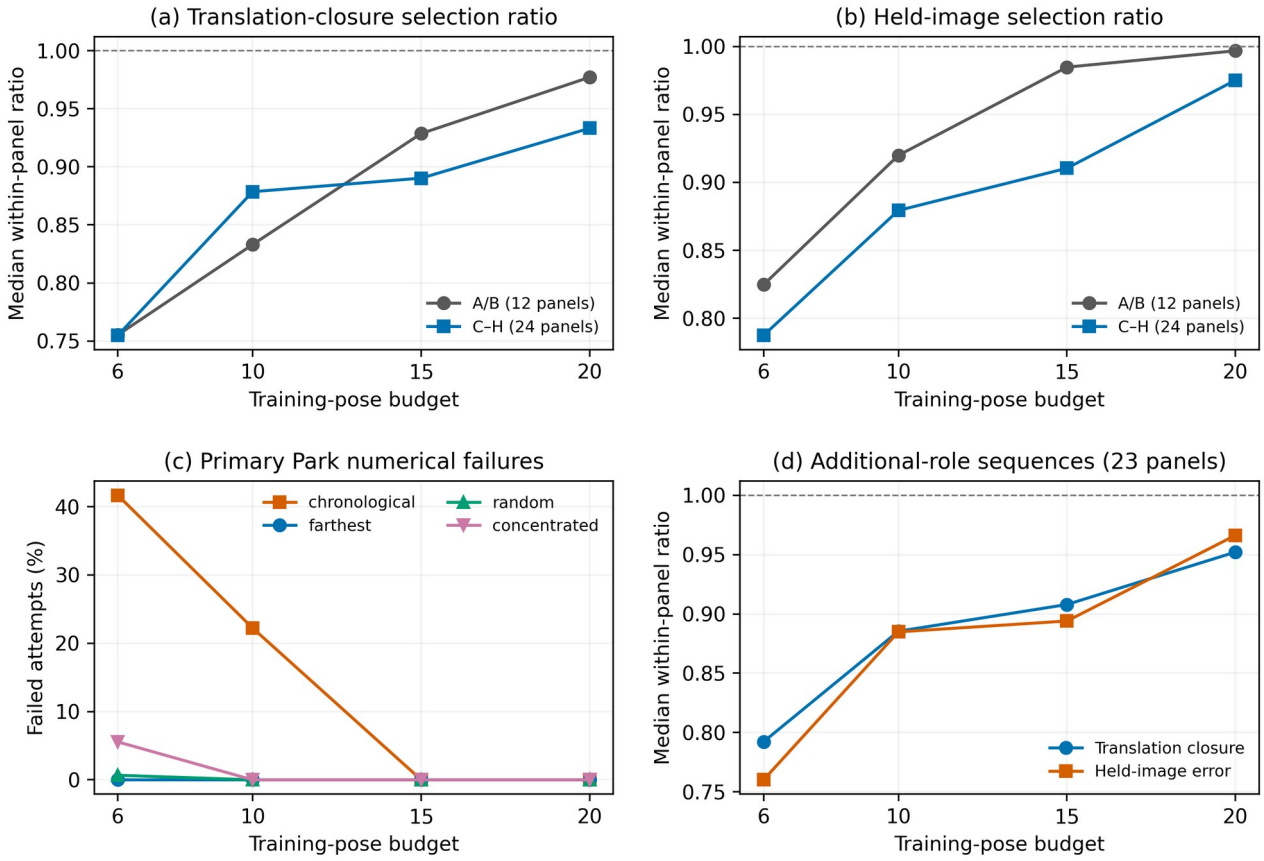


Figure 4. Pose-budget dependence and numerical failures. Panels (a,b) compare median within-panel ratios for A/B and C–H; (c) retains all attempted primary Park fits; (d) reports the 23 additional-role sequences separately. A finite negative-depth estimate is additionally flagged in the numerical tables. Source: author's measurements and calculations.

4.3 Matched solver and pixel-objective effects

The 2,808-attempt experiment exposes failures that depend on both solver and selection policy (Figure 6). Across 468 ten-pose subsets, valid results number 92 for Tsai, 344 for Andreff, 459 for Park, 459 for Horaud, 462 for Daniilidis and 459 for Park-initialized pixel refinement. Tsai has no valid body or wrist result; Andreff has no valid head result. These counts refer to repeated computational fits on measured poses, not additional physical trials.

On the 36 farthest subsets valid for Park, Horaud and Daniilidis, median held-image errors were 0.432, 0.432 and 0.416 pixels, respectively. Paired Daniilidis/Park ratios had a median of 0.988 and range of 0.875–1.330. Under concentrated selection, their paired median ratio was 1.012 (Table S5). Thus the solver comparison also depended on pose arrangement; the lowest conditional median did not imply uniformly lower error.

Chronological ten-pose subsets are particularly restrictive. Park yields 28/36 finite estimates but only 27/36 positive-depth estimates. Horaud is finite in 36/36 but positive-depth in 27/36; Daniilidis is finite in 33/36 and positive-depth in 30/36. Pixel refinement returns finite estimates for 28/36, converges for 27/36 and has 27/36 valid results. In contrast, Park, Horaud, Daniilidis and pixel refinement each yield valid estimates for all 36 farthest subsets and all 360 random subsets. Finite transforms alone are therefore insufficient for acceptance.

Within matched valid Park/pixel pairs, median held-image reductions are 22.3% for farthest selection (36 pairs) and 29.7% for random selection (360). Reductions are 52.5% for concentrated selection (36) and 49.4% for chronological selection (27). Held-image errors improve in 35/36, 354/360, 35/36 and 27/27 pairs, respectively. These paired changes quantify the objective effect without comparing different successful solver subsets.

The relative coverage advantage narrows after pixel fitting (Figure 5b). Using the same farthest subset and ten random draws per panel, the median image-error ratio changes from 0.911 under Park to 0.993 after pixel

refinement. The refined group-balanced ratio is 1.001 and 22/36 panel ratios are below one. This is not an equivalence test. After pixel fitting, farthest selection still gives lower error than concentrated selection in 29/36 matched panels, with a median ratio of 0.870. Against valid chronological subsets, it gives lower error in 25/27 panels, with a median ratio of 0.882. The weak farthest/random contrast therefore does not imply that all pose arrangements are interchangeable.

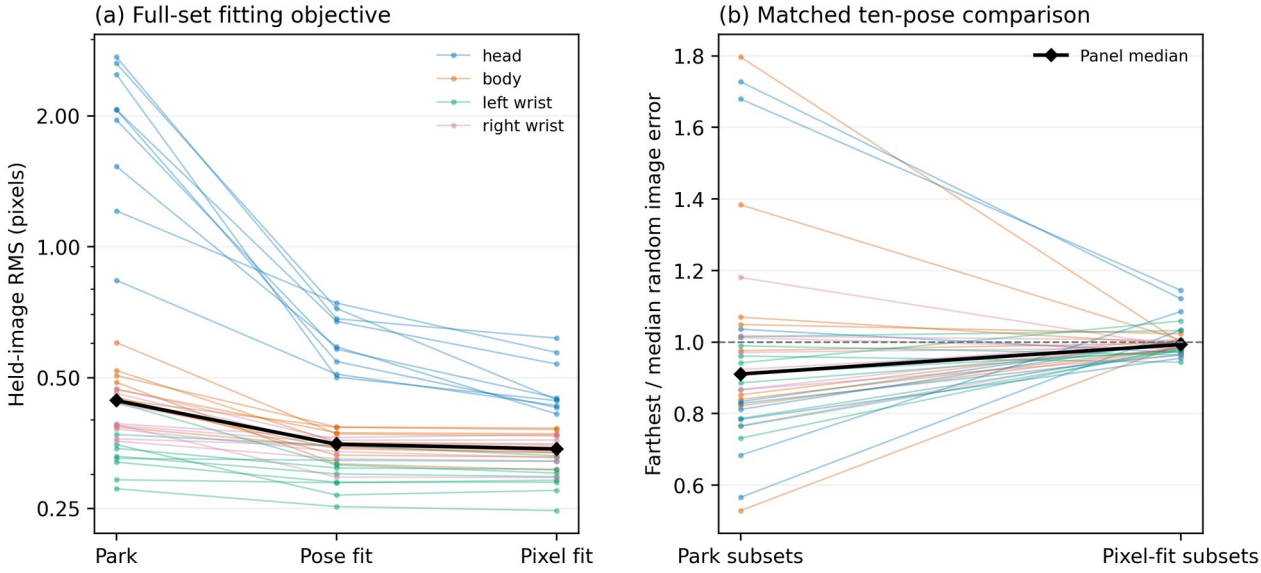


Figure 5. Objective dependence on a common held-image endpoint. (a) Full candidate sets under Park, pose-closure refinement and pixel refinement; thin lines are panels and black lines are medians. (b) Farthest/random ratios using the same ten-pose farthest subset and ten saved random draws per panel; the median changes from 0.911 to 0.993. Random draws are shared between objectives. Source: author's measurements and calculations.

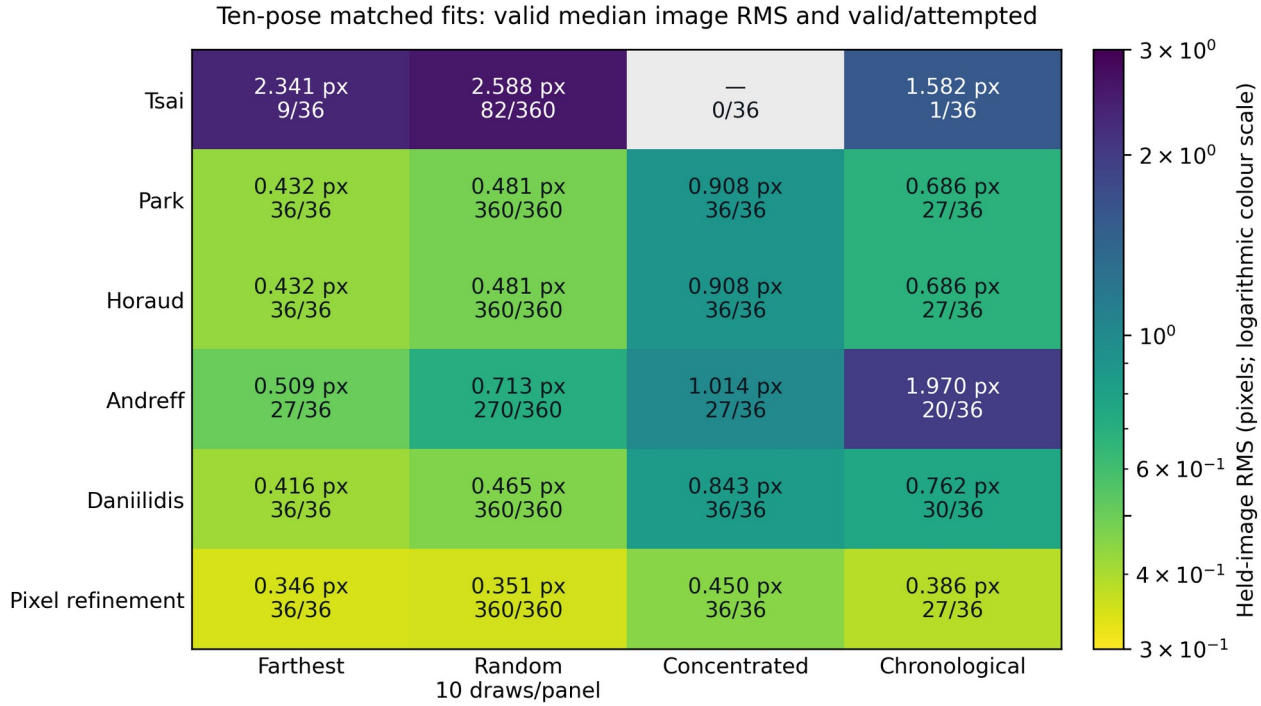


Figure 6. Matched ten-pose solver/objective comparison. Each cell gives the median held-image RMS among valid results and valid/attempted counts. Validity requires finite output, convergence where applicable, and positive target-corner depth in training and validation. All methods use identical subset membership. Different valid subsets preclude a universal ranking from the cell medians. Source: author's measurements and calculations.

4.4 Sensitivity and transfer conditions

Omitting one training pose changes X by up to 8.554 mm in translation and 0.463° in rotation. These shifts measure finite-design sensitivity, not uncertainty in the true camera mounting. Five specified geometric scale combinations give main-sample ten-pose median closure ratios of 0.847–0.883.

In the eight cross-date transfers, head, body and left-wrist image-error ratios relative to a destination-specific fit are 0.99–1.14. The two right-wrist directions give 89.03 and 97.84, coinciding with a 179.75° difference between fitted X rotations. The comparison cannot distinguish mounting changes, image-coordinate changes and kinematic configuration effects. It consequently supplies a configuration-continuity warning, not a measurement of mechanical drift.

The 52 matching-configuration same-day transfers have a median ratio of 1.005 and range 0.922–1.199. Five further head sequences give larger full-set Park errors, 8.818–9.957 mm and 5.487–6.047 pixels, reduced to 0.617–0.718 pixels after pixel fitting. These repeated single-role sequences remain separate from the main 36 panels and limit extrapolation of its residual range.

5 Discussion

The main finding is an interaction between pose arrangement and fitting objective. With Park estimation, farthest-point selection often improves local held-pose and held-image consistency at a small pose budget. With the same ten-pose subsets and a joint pixel fit, the farthest/random image comparison approaches unity. Larger paired reductions for concentrated and chronological subsets explain why changing the objective narrows the apparent coverage advantage. At the same time, failures under chronological selection show that optimizer choice cannot compensate for every deficient motion set.

This result qualifies the practical role of geometric selection. Farthest-point sampling is an inexpensive way to avoid concentrated candidate subsets, and it needs only training robot poses. It does not model corner uncertainty or acquire views beyond the candidate pool, as active approaches aim to do (Yang *et al.*, 2023; Li *et al.*, 2026). For the tested platform and local validation region, the results support preserving rotational diversity and testing solver validity at the intended budget. Training image refinement and separate-pose assessment using closure and projection then provide complementary checks. The experiments do not determine a universal minimum number of poses.

Closure and image error measure related but different aspects of the same calibration chain. Closure combines estimated target poses with the robot model; the image endpoint tests whether one shared X/Z projection explains omitted corners. Unlike independently fitting PnP to every validation image, that projection cannot adjust extrinsics for individual images. However, both endpoints retain errors in intrinsics, kinematics, target geometry and time association. Their numerical improvement should not be translated directly into improved end-effector accuracy, consistent with the need for application-level calibration evidence (Bai *et al.*, 2020; Wang *et al.*, 2025).

Implementation and acquisition conditions are part of the result. The small body/wrist rotations trigger a specific Tsai pair filter, and the two-cell target border defines the detector-compatible corner convention. Similar named solvers or fiducial families can therefore give different outcomes if these details are omitted. Likewise, the transfer discrepancy shows why a camera-to-support estimate should be validated again when its mounting or coordinate configuration is uncertain. The same-day analysis narrows some recorded differences but does not isolate a physical cause.

The four camera roles share a platform family but are calibrated separately. Extending these findings to camera-to-camera registration would require synchronized shared-target observations and an independent relative-transform assessment of the kind motivated by Allegro *et al.* (2024). The present evidence supports local camera-to-support consistency under the stated target, kinematics and projection model.

6 Limitations

The experiments do not include external metrology, independently measured target scale or flatness, controlled remounting, or a manipulation/manufacturing task test. The nominal target dimensions set the

metric scale. A small pixel residual can coexist with biased X when kinematic or intrinsic errors are absorbed into the fitted transforms. Exposure timing, image preprocessing and compression can also influence corner localization.

The achieved motions are role-dependent and validation poses lie close to training poses. Mechanism, camera role and trajectory family covary, preventing causal attribution to a single coverage descriptor. Nine sessions and eight acquisition groups do not supply independent samples for every image, panel, subset or directed transfer. Group D lacks a session-stored robot identifier; its exclusion sensitivity reduces that particular uncertainty.

The sample is conditioned on completed, retained camera sequences and accepted acquisition windows. It excludes attempts without usable measurements, so numerical solver success must not be interpreted as end-to-end collection success. Validation was separate during collection but accessible to the calibration workflow, and the secondary matched experiment was informed by the primary results. The findings are consequently exploratory beyond the specified settings. A new, broader validation session with independent target metrology would test whether the observed local consistency transfers to absolute positioning.

The numerical package supports solver, selection and objective comparisons from measured corners and transforms. Repeating image detection additionally requires the original images and compatible detector; an exact acquisition-date software environment is not established for every session. No controlled comparison of camera mechanisms, active pose acquisition, synchronized network calibration or uncertainty-weighted pixel objectives is claimed.

7 Conclusions

Experiments on four robot-camera roles show that pose coverage, solver validity and optimization objective must be assessed together. At ten poses, the main-sample farthest/random median ratio is 0.854 for translation closure and 0.912 for held-image error under Park fitting. In a matched ten-random-draw comparison, the image ratio changes from 0.911 to 0.993 after training-only pixel refinement. All 36 full-set pixel fits converge with positive depth and achieve 0.247–0.616 pixel RMS on the eight local validation poses per panel.

The attained effect is improved local model consistency under fixed intrinsics, target geometry and robot kinematics. It does not establish absolute extrinsic accuracy or task positioning performance. Equal pose budgets, paired objective comparisons, explicit failures and separate validation poses provide a practical basis for assessing these calibrations within their demonstrated operating conditions.

Data and code availability

Measured transforms, corner coordinates, intrinsics, pose/frame identifiers and numerical reanalysis code are available at https://github.com/jensenzhou47/robot-calibration-pose-coverage-data (version 1, commit 8ee9ffe1819ccd5096c912d82fb4c163184d4386). Version 1 contains 36 main panels, 23 additional-role panels and five further head sequences. The numerical observations are also supplied as plain-text supplementary files. Supplementary_material.docx gives per-panel results and processing details. Supplementary_matched_benchmark_code.txt and Supplementary_matched_benchmark_results.txt provide the additional ten-pose benchmark, its fixed protocol and 2,808 results. These additions are separate from the version-1 repository commit. These resources reproduce numerical fitting and comparisons. Full image redetection additionally requires the original images and compatible detector. No repository DOI is claimed.

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